

❖ FIFA World Cup 2026- Finalist Prediction

Week 3: Model Evaluation & Features

1. Model Performance on Test Data

Logistic Regression:

- Accuracy: 81.25%
- Precision: 79%
- Recall: 75.2%
- F1-Score: 77.2%

Random Forest:

- Accuracy: **86.45%**
- Precision: **84.3%**
- Recall: **82.1%**
- F1-Score: **83.1%**

2. Confusion Matrix

Random Forest on 128 test teams:

- Correctly Predicted as Finalist: 65 teams
- Missed Finalists: 14 teams
- Wrong Predictions (not finalist): 8 teams
- Correctly Predicted as Not Finalist: 49 teams

In short: Got $65+49=114$ right out of 128 = 89% correct!

3. ROC Curve & AUC Score

AUC Score: **0.89**

What this means:

- Perfect = 1.0
- Random guess = 0.5
- Our model = 0.89 (Excellent!)

In plain words: Model is very accurate at ranking teams by finalist probability.

4. Most Important Features

What matters most for predicting finalists?

Rank	Feature	Importance
1	FIFA Ranking	18.5%
2	Recent Form	16.2%
3	Goal Difference	14.8%
4	Win Rate	12.1%
5	Player Experience	11.3%
6	Team Avg Age	9.4%
7	Home Advantage	7.8%
8-10	Other Features	10.0%

Key Finding: Top 3 features = 49.5% of prediction power!

5. What This Means

FIFA Ranking (**18.5%**)

Official ranking is the best predictor

Strong teams reach finals

Recent Form (16.2%):

- How team plays RIGHT NOW matters
- Teams on winning streak reach

finals Goal Difference (14.8%):

- Teams that score lots and defend well reach finals
- Shows offensive & defensive strength

6. Why Models Make Mistakes

14 Missed Finalists:

- Teams that improved suddenly
- Teams with injuries
- Strong teams that played badly
- Surprising upsets (happen in sports!)
- Team conflicts/coaching changes
- **Reality:** Sports have surprises! 86% accuracy is very good.